Azfar Bilal

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Computer Vision & Edge AI Engineer. I build real-time detection, tracking and multi-camera systems and make them fast enough to ship with TensorRT, custom CUDA kernels, NVIDIA Jetson and AWS. **Open to relocating to the US.**

Education

FAST National University of Computer and Emerging Sciences (NUCES) Aug. 2022 – Jun. 2026
Bachelor of Science in Computer Science *Islamabad, Pakistan*

- **Dean's Honour List** (academic excellence). Coursework: Deep Learning, MLOps, Generative AI & LLMs, Artificial Intelligence, Data Structures, Operating Systems.

Experience

AI / Computer Vision Engineer Dec. 2025 – Present
SovaVis *Islamabad, Pakistan*

- Built **AwareCam**, a video analytics platform running **10 to 12 concurrent RTSP feeds on one GPU at 100+ FPS** with **under 100 ms** latency and zone- and schedule-aware alerts (YOLO11, TensorRT, ByteTrack, ALPR).
- Engineered **LotIQ**, a multi-tenant AWS parking-violation SaaS with **cross-camera vehicle re-identification** (OSNet) and automated towing; its GPU pipeline idles at 1 FPS and scales to 8 FPS only on candidate violations.
- Built an **offline face-recognition door access system** for a Dubai client on **NVIDIA Jetson Orin Nano**, driving relay-actuated locks over GPIO.
- Shipped **Taziza**, a smart-fridge checkout that bills in real time from line-crossing across two cameras, with QR checks for look-alike products; rebuilt it in **TensorRT** for real-time throughput.
- Optimized models (TensorRT FP16/INT8, ONNX Runtime) and deployed inference services with Docker, FastAPI, PostgreSQL, AWS (Lambda, ECS, SQS, S3) and CI/CD (GitHub Actions, MLflow).

Projects

Neural Network Acceleration with CUDA | *CUDA C++, TensorRT, ONNX, OpenACC* May 2025

- **12× inference speedup** with no accuracy loss via custom CUDA kernels and an ONNX-to-TensorRT export path; built a multi-GPU training pipeline with CUDA streams and unified memory.

RAG + Agentic Detection System | *LangChain, FAISS, OpenAI, YOLOv8, DeepSORT, FastAPI* 2025

- An LLM agent queries live detections through **tool-calling**, raises alerts and writes summaries over a RAG knowledge base; **95%** recognition accuracy at **45+ FPS** on AWS EC2.

Rubric.ai, AI Exam Evaluation Platform | *Next.js, React, OpenAI, OCR, Node.js* 2025

- End-to-end EdTech product (handwriting OCR, LLM grading, instant feedback); launched to **200+ users in the first week** with paying users.

Realistic Logo Compositor | *OpenCV, Homography, Depth Estimation, Poisson Blending* 2025

- Placed logos onto real surfaces in photos with correct perspective, lighting and occlusion using classical CV (homography warp, depth-guided shading, Poisson blending).

Technical Skills

Computer Vision: YOLOv8/YOLO11, OpenCV, ByteTrack, DeepSORT, OSNet re-ID, ALPR/OCR, face recognition, multi-camera tracking

Inference & GPU: TensorRT (FP16/INT8), ONNX Runtime, CUDA C++ (custom kernels), CUDA streams, quantization, multi-GPU

Edge & Streaming: NVIDIA Jetson Orin, GPIO/relay control, RTSP/WebRTC, GStreamer, FFmpeg, MediaMTX

ML & LLM: PyTorch, TensorFlow, OpenAI API, LangChain, LangGraph, RAG, AI agents, FAISS, Pinecone

Cloud & Backend: AWS (EC2, S3, Lambda, ECS, SQS), FastAPI, Node.js, PostgreSQL, Docker, Kubernetes, MLflow

Languages: Python, C/C++, CUDA C++, JavaScript/TypeScript, SQL, Bash